

Project Design Phase

Solution Architecture

Date	7 July 2026
Roll Number	23A51A4265
Project Name	Leave Approval System
Maximum Marks	4 Marks

Solution Architecture: Leave Approval System

Overview: The Leave Approval System bridges the gap between organizational leave management problems and technology solutions by automating the entire workflow through ServiceNow. It ensures accurate validation, streamlined approvals, and transparent communication between employees and managers.

Key Goals:

- Provide the best technical solution to automate leave requests and approvals.
- Define the structure and behavior of the system for stakeholders.
- Ensure features like validation, workflow automation, notifications, and reporting are clearly specified.
- Deliver a scalable, secure, and efficient solution on the ServiceNow cloud platform.

◆ Architecture Components

1. Frontend (Employee Interaction)

- ServiceNow Portal & UI Forms for submitting leave requests.
- REST API for communication with backend logic.
- Client-side validation to prevent incorrect date entries.

2. Backend Logic

- Validation engine ensures end date $\geq$ start date.
- Flow Designer routes requests to managers.
- Business rules handle workflow consistency.

3. Approval Workflow

- Manager reviews requests and approves/rejects.
- Decision recorded in system and linked to employee dashboard.

4. Database & Storage

- ServiceNow tables store leave records and approval history.
- File storage for attachments (medical certificates, etc.).

5. Notification Service

- Sends approval/rejection notifications via email/SMS.
- Integrated with external APIs like Twilio or SMTP.

6. Cloud Infrastructure

- Hosted on ServiceNow Cloud Platform.
- Ensures scalability, availability, and secure access.

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